Margaret Wozniak

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Education

University of Michigan, College of Engineering **Ann Arbor, MI**

M.S.E Computer Engineering, Robotics Concentration | Expected May 2028

B.S.E. Computer Science Engineering, Data Science Minor | August 2023 - December 2026

- **GPA**: 3.767/4 .00
- **Honors/Awards:** Dean's List and University Honors: 6 consecutive semesters
- **Relevant Coursework:** Embedded Systems Design, Computer Vision, Parallel Programming, Autonomous Controls
- **Activities & Organizations:** University of Michigan Club Field Hockey; BLUElab Data; SAXA Senior Honor Society

Experience

Consumers Energy **Jackson, MI**

Database Administration Intern May 2026 - August 2026

- Engineered a KQL-driven Azure monitoring solution with enterprise-wide Log Analytics Workspaces, implementing time-series analytics and statistical workflows to automate anomaly detection across 92 cloud database instances.
- Automated infrastructure health reporting and risk mitigation systems, reducing manual administrative workloads by 10% while increasing critical alert detection by 32% and cutting false alerts by 17%.

Beach Internals **Remote**

AI Integration Engineering Intern June 2025 - September 2025

- Built a generative AI solution to automate volleyball analytics, transforming raw match data into targeted, easy to read formats that effectively supported both technical and non-technical stakeholders.
- Embedded automated summaries into production platforms using a custom Anvil Uplink pipeline, successfully reducing turnaround times and boosting overall operational efficiency by 25%.

Research & Projects

Beyond Visual Spectrum Sensing for Autonomous Robotics **Ann Arbor, MI**

University of Michigan ROB 490: Directed Study July 2026 - Present

- Developed and evaluated multimodal navigation algorithms for autonomous robotic systems, integrating visual, thermal, and depth-guided sensor pipelines into core perception and navigation frameworks.
- Engineered hardware and software architectures for sensor data acquisition while authoring comprehensive technical documentation, including lab guides, instructor resources, and technical reports.

Weight Controlled Labyrinth **Ann Arbor, MI**

University of Michigan EECS 373: Embedded Systems January 2026 – April 2026

- Developed a UART-based wireless communication protocol between a fitness board and maze controller using ESP32-C6 modules, including transmission handling and UART overrun error recovery for reliable data exchange.
- Designed the maze-side embedded software and hardware architecture, implementing control logic and state machines while preserving motor control and beam-break sensor functionality.
- Performed system integration by wiring, assembling, and validating embedded electronics and mechanical components to produce a functional weight-controlled labyrinth.

Leadership

Michigan Club Field Hockey **Ann Arbor, MI**

Director of Operations, Captain August 2025 - Present

- Led the team to a 3rd place national finish while managing all logistics and operations for a 22 player roster.
- Boosted team retention by over 40% by designing new training programs focused on team bonding and skill development while supporting and encouraging a collaborative team culture.

University of Michigan BLUElab Data **Ann Arbor, MI**

Machine Learning and Data Forecasting Lead April 2024 - May 2026

- Directed a cross-functional engineering team through the development and deployment of a hybrid machine learning model, mentoring members across disciplines to optimize demand predictions and reduce MDining waste by 10%.
- Facilitated cross-departmental collaboration with University executives and kitchen staff, guiding the implementation of real-time forecasting workflows that transformed daily kitchen operations and minimized excess costs.

Technical Skills

- **Programming Languages:** C++, C, C#, Java, KQL, SQL, R, Powershell, Julia, MATLAB, CUDA, HTML, JSON, Javascript
- **Cloud and Platforms:** Azure, Github, Linux, AWS (EC2), Anvil, Google Colab, Terraform, Bicep, PowerBI, VScode